Generating Chest Radiographs with a Convolutional Variational Autoencoder

ECE 285 Homework 1

Abstract

We train one convolutional variational autoencoder (VAE) on the complete Chest X-Ray Images (Pneumonia) training set and draw unconditional samples from its Gaussian prior. We report training behavior, example radiographs, Inception Score (IS), and an explicitly identified 1000-feature Fréchet Inception Distance (FID-1000). The generated images capture coarse chest structure but blur fine anatomy; these natural-image metrics do not establish diagnostic quality.

1 Data and method

We use the Kaggle pediatric chest X-ray dataset [1]. The original training split is stratified into 4,694 training and 522 validation images (seed 285); the separate 624-image test split is reserved for evaluation. One VAE covers both NORMAL and PNEUMONIA images. Images are converted to grayscale, resized to 64×64, and scaled to $[0, 1]$.

Four stride-two convolutional encoder layers produce $4 \times 4 \times 128$ features. Linear heads parameterize a 32-dimensional diagonal Gaussian posterior; a reparameterized sample enters a linear layer and four transposed convolutions ending in a sigmoid. The model has 543,649 trainable parameters. Following Kingma and Welling [2], we minimize per-image pixelwise binary cross-entropy (summed over pixels) plus $\beta D_{\mathrm{KL}}[q_\phi(z \mid x) \| \mathcal{N}(0, I)]$, with $\beta = 1$. At generation time, $z \sim \mathcal{N}(0, I)$; no test image is supplied.

2 Training

We train with Adam (learning rate 10^{-3}), batch size 64, for 20 epochs on CPU. The checkpoint is selected by minimum validation negative ELBO. The test split does not enter training or checkpoint selection. At epoch 20, the best validation loss is 2,388.10 nats/image (reconstruction 2,365.96; KL 22.14); the final training loss is 2,397.53. Figure 1 shows the loss trend.

3 Generative evaluation

We compare 256 independent prior samples with 256 randomly chosen official test images. For Ignite's IS and FID implementation [3], each grayscale image is repeated to RGB, resized to 299×299, and evaluated by ImageNet-pretrained Inception-v3. The scores are IS = 1.779 and FID-1000 = 0.177. FID-1000 uses the model's 1000-class probabilities as features, as in the cited Ignite tutorial; it is not comparable to the usual pool3/2048-feature FID [4]. The sample count is small, and an ImageNet classifier is not calibrated to radiographic anatomy.

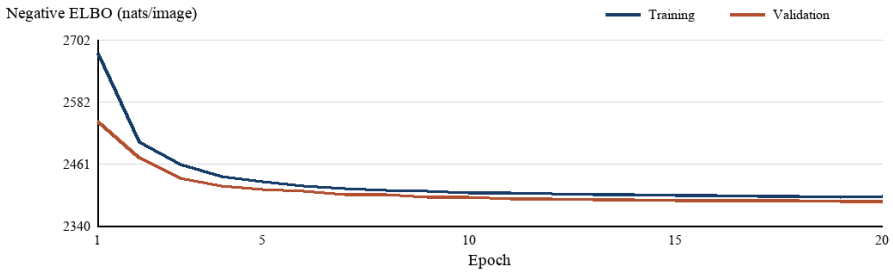
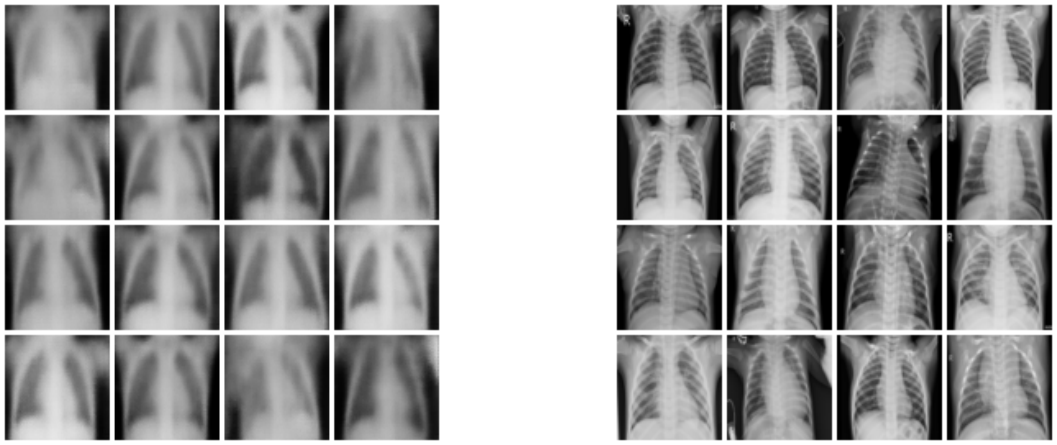


Figure 1: Mean negative ELBO in nats/image on training and validation splits; lower is better. The selected checkpoint is epoch 20.



(a) Prior samples (b) Held-out test images

Figure 2: Generated and real 64×64 grayscale radiographs displayed at the same scale. The model recovers coarse chest outlines but loses rib and lung texture.

4 Discussion

Figure 2 shows visibly smoother generated images than real test images. Downsampling removes fine lung detail, and pixelwise binary cross-entropy with a compact Gaussian latent model can favor smooth outputs. The training set is class-imbalanced; this unconditional model cannot select pathology. IS and FID-1000 describe only this specific 256-sample experiment, not clinical realism or diagnostic performance. Higher resolution and radiology-specific evaluation would be useful next steps.

References

[1] P. Mooney. Chest X-Ray Images (Pneumonia). Kaggle, 2018. `https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia`.

[2] D. P. Kingma and M. Welling. Auto-Encoding Variational Bayes. ICLR, 2014.

[3] PyTorch-Ignite. GAN evaluation using FID and IS. `https://pytorch-ignite.ai/blog/gan-evaluation-with-fid-and-is/`.

[4] M. Heusel et al. GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium. NeurIPS, 2017.